Haihao Lin

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Research interests: VLA post-training and representation stability; real-robot learning; cross-task generalization; deployment-aware model adaptation.

Education

University of Chinese Academy of Sciences (UCAS) Sep. 2024–Jun. 2027 (expected)
M.S. in Computer Science (Artificial Intelligence). Advisor: Prof. Xiangsheng Huang.

Jiangsu University Sep. 2020–Jun. 2024
B.E. in Computer Science and Technology. GPA rank: **1/171**.

Research

FiberTune: Preserving Action-Fiber Visual Residuals in Vision-Language-Action Fine-Tuning
First and corresponding author | Accepted at CoRL 2026 | arXiv:2606.08653 | Project page: fibertune.github.io

- FiberTune addresses a limitation of action-supervised VLA fine-tuning: only feature directions that change predicted actions are constrained, allowing visual structure shared across action-equivalent states to degrade. During training, an action probe estimates and filters action-predictive components; the resulting visual residual is aligned to a frozen visual teacher and regularized with an effective-rank prior to prevent collapse into a few dominant directions. The deployed policy and inference cost remain unchanged.
- Validated FiberTune across $\pi 0.5$ and OpenVLA-OFT, the CALVIN and LIBERO benchmarks, and a real-world SO-101 task. It outperformed task-loss-only fine-tuning in all six controlled settings; CALVIN ABC$\rightarrow$D SR(5) increased from 61.4% to 72.1%, and SO-101 task success from 72.7% to 78.1%.
- Built the SO-101 data-collection and OOD evaluation pipeline; conducted controlled training, evaluation, and representation diagnostics with $\pi 0.5$ on CALVIN, LIBERO, and SO-101, and with OpenVLA-OFT on LIBERO.

Honors and Competitions

- National Scholarship (twice); Jiangsu Provincial Merit Student; Jiangsu University President's Medal; Outstanding Graduate; Outstanding Undergraduate Thesis.
- Three silver medals at ICPC Asia Regional Contests; highest-ranked bronze-medal team at the 2024 ICPC Asia-East Continent Final; two First Prize awards at the Lanqiao Cup National Finals, one each in C/C++ and Python (Python: 1st in Jiangsu, 9th nationally).

Current Research

- Post-training, representation, and control interfaces for embodied foundation models, spanning VLAs and WAMs, with an emphasis on cross-task generalization and real-robot learning.

Robotic Systems

ROS2 Systems Engineering for a Commercial Cleaning Robot Mar. 2026–Aug. 2026

- Designed and implemented autonomous docking for charging, cliff detection and avoidance, and a corresponding docking simulation system for a commercial cleaning robot, covering sensor selection and placement, charger communication, state monitoring, and docking control; performed simulation-based parameter tuning and stress testing, ROS2 deployment on the real robot, and system-level closed-loop validation.
- Implemented the first version of the coverage-path planner, adapted an external algorithm, and completed real-robot integration testing; optimized wake-word detection and speech recognition for the robot's noisy operating environment; completed function integration and system-level state-machine commissioning.

Intelligent Logistics Robot System Design and Prototype Validation Aug. 2025–Jan. 2026

- For logistics applications, built an xArm + RGB-D tabletop manipulation prototype and deployed it on Jetson, covering hand-eye calibration, suction/gripper integration, and Web API integration.

Additional Research and Engineering

Task Planning Algorithm Research (Confidential Project) Mar. 2024–Oct. 2024

- Implemented multiple planning and scheduling algorithms, randomized data generators for multiple scenarios, and a complete test framework; consolidated the team's technical proposal and bid materials, prepared the presentation deck, and delivered the on-site presentation.

Property Prediction and Screening for an AI-Enabled Pharmaceutical Platform Sep. 2023–Jun. 2024

- Completed data processing and model replication for the large-molecule workstream of an AI-enabled pharmaceutical platform; developed a property-prediction and screening system; contributed to service API implementation and platform integration. The resulting thesis was recognized by the university as an Outstanding Undergraduate Thesis.

MaaAssistantArknights **Open-Source Automation Project** Jun. 2022–Apr. 2026 | GitHub 20k+ stars

- As an early key contributor to MaaAssistantArknights, developed multiple C++ core modules and WPF GUI features, and implemented and iteratively improved a task-planning algorithm for mutual-exclusion constraints.

Skills

Programming and frameworks: Python, PyTorch, LeRobot, and RLinf.
Embodied models: the π family (including π0.5), the OpenVLA family (including OpenVLA-OFT), SmolVLA, and FastWAM.
Robotics and systems: ROS2, C++/CMake, OpenCV, RealSense/RGB-D, xArm SDK, Jetson/RK, and Linux/Docker.